

Final R&D Claim Report

ABC Ltd

UTR: 0123456789

Research & Development Tax Credit

Accounting period ended
31 December 2021

Fully Flexible Simulation Rapid Build Algorithm

Technological advances

The ABC team developed a proprietary simulation rapid build algorithm that enables clients to build their own simulation models with only a minimal level of training.

The simulations are an accurate representation of real-world processes and have been optimized for training and education purposes. To that end, the output standard is one that allows deployment onto everyday laptop, tablet, and smartphone devices, as either web-based or app-style formats.

A typical complex system, such as an aircraft's hydraulics system can be built within 30 minutes using the SDK holding the right graphics. Once built, the simulation can then be used for familiarization, basic or advanced training in both normal and abnormal operations.

The scope of the subject matter is only limited by the graphics contained with the SDK – and hence can be changed to incorporate an almost unlimited range of topics. However, initial development of the SDK has been limited to the aerospace sector – not least because simple modifications to the SDK will then allow a much wider range of topics to be incorporated.

Technological uncertainties

During this financial period, the team faced several challenges, particularly when having to ensure that the algorithm was flexible enough to function across an almost infinite range of topics, including mechanical, biological and virtual systems. After creating the algorithm to work with such diversity it was also vital this be translated onto the graphical interfaces to provide the user with full instruction. Furthermore, due to the growing demand of 3D/Virtual Reality technologies ABC made it their aim to support these graphical technologies as well as providing flexibility/scalability for any future developments of this type of technology.

During this phase of research and development, the ABC team went through several steps of trial-and-error iterations to continually evaluate the limiting functionality and accuracy of the algorithm. Testing was undertaken on a variety of systems, and it was initially found that the algorithms wouldn't create satisfactory results and were often wildly inaccurate. The team then had to regroup and uncover the reasons behind these imprecisions and rebuild. This aspect of building the technology is ongoing and will vary between processes it's trying to replicate.

Once this phase produced satisfactory results, the team used further trial and error to translate the simulations into 3D and virtual reality visuals. Because these technologies are evolving rapidly and vary widely, ABC must continue checking any developments to ensure full compatibility. Accuracy is essential when modelling the granular details of complex processes.

CONFIDENTIAL | This document contains proprietary information of **ABC Ltd**. Do not disclose or distribute externally unless explicitly requested or authorized on a strict need-to-know basis.